

Temperature-switchable thermal barriers against thermal runaway propagation

A one-dimensional finite-volume study

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github.com/adityameenak/switchable-barrier-runaway

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Abstract

A thermal barrier between lithium-ion cells faces a conflict. A conductive barrier spreads heat to the cooling system in normal operation but carries a runaway cell's heat to its neighbours. An insulating barrier blocks propagation but traps heat during normal use. A barrier whose conductivity drops sharply above a switch temperature could, in principle, do both. This report quantifies that tradeoff with a one-dimensional transient finite-volume model of a five-cell stack, using single-step Arrhenius self-heating, a smooth reversible switch law $k(T)$ and convective cooling at the stack ends. With illustrative parameters, a 2 mm switchable barrier holds the normal-operation peak temperature at the conductive barrier's 38.0 °C (against 64.5 °C for an aerogel-like insulator) while confining runaway to the trigger cell, whereas a conductive barrier lets runaway spread to all five cells within about a minute. A sweep of 361 barrier designs shows that containment is governed by a single quantity, the hot-state barrier conductance k_{off}/L , which must stay below about 32 W m⁻² K⁻¹ for the chosen end cooling. A switch-temperature sweep finds a window of about 38–175 °C in which the switch both cools and protects. The solver conserves energy to round-off and is verified against analytical, ODE-solver and convergence tests. The convergence study also showed that onset-fitted Arrhenius kinetics, extrapolated to burning-cell temperatures, produce a reaction front too thin for any practical mesh to resolve; a rate ceiling restores mesh convergence. All parameters are order-of-magnitude placeholders, so the conclusions are qualitative.

1 Introduction

Thermal runaway in a lithium-ion cell is a self-accelerating sequence of exothermic decomposition reactions that can drive the cell to several hundred degrees and lead to fire [1–3]. In a module, the heat released by one failing cell can trigger runaway in its neighbours, as multi-cell abuse experiments have shown repeatedly [4, 5]. Preventing this cell-to-cell propagation is a central goal of battery-pack safety design. The usual approach places a thermal barrier between cells, which creates a design conflict:

- a *conductive* barrier helps move the heat cells generate in normal operation toward the cooling system, but also passes a runaway cell's heat directly to its neighbour;
- an *insulating* barrier, for example an aerogel sheet, blocks propagation but raises cell temperatures in everyday use, which accelerates ageing.

Temperature-responsive barriers and thermal switches, whose conductivity changes with temperature, are an active research topic as a way out of this conflict [6, 7].

Related work. Thermal abuse of single cells is commonly modelled by coupling heat conduction to Arrhenius kinetics for a set of decomposition reactions, an approach established for cylindrical and prismatic cells [8], applied to abuse behaviour of high-power cells [9], extended to three dimensions [10] and combined with experiments [11]. Propagation models extend this to packs and modules, with lumped [12] and three-dimensional descriptions [13] and analyses of the heat-transfer paths between cells [14]. On the mitigation side, studies have examined phase-change composites [15], aerogel-based insulation [16], model-based optimisation of heat-absorbing interstitial barriers [17] and polymer barrier materials [18]. These barriers are static: their conductivity does not respond to the operating state. Thermal switches, devices whose conductance changes with temperature or another stimulus, are reviewed in general in [19] and for batteries in [7], and a temperature-responsive regulator for battery modules has been demonstrated experimentally [6].

This work. This report complements that work with a deliberately minimal model that isolates the conductivity-switching function of a barrier, and maps which switch properties are needed for containment without penalising normal operation. It asks three questions:

1. Does an idealised switchable barrier deliver both the conductor’s normal-use temperature and the insulator’s containment?
2. What hot-state conductivity and thickness does it need?
3. How should its switch temperature be chosen?

The emphasis is on a transparent, verified model rather than quantitative prediction. Every material and kinetic parameter is an illustrative, order-of-magnitude value and has not been fitted to any particular cell or barrier.

2 Model

2.1 Geometry and governing equations

The stack is five cells of thickness $L_c = 10$ mm separated by four barriers of thickness L (default 2 mm), modelled in one dimension through the stack thickness. Temperature obeys transient conduction with a temperature-dependent conductivity and volumetric sources,

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k(T) \frac{\partial T}{\partial x} \right) + q_{\text{rxn}} + q_{\text{gen}}, \quad (1)$$

with convective boundary conditions at both ends of the stack,

$$-k \frac{\partial T}{\partial x} \Big|_{x=0} = h (T_\infty - T), \quad k \frac{\partial T}{\partial x} \Big|_{x=L_{\text{stack}}} = h (T_\infty - T). \quad (2)$$

2.2 Cell self-heating

Each cell node carries a lumped reaction progress $\alpha \in [0, 1]$ governed by single-step kinetics, a simplification of the multi-reaction schemes used in detailed abuse models [8–10]:

$$\frac{d\alpha}{dt} = k_{\text{eff}}(T) (1 - \alpha), \quad q_{\text{rxn}} = \Delta H_v \frac{d\alpha}{dt}, \quad \Delta H_v = \rho c_p \Delta T_{\text{ad}}, \quad (3)$$

so that a cell reacting adiabatically heats by exactly ΔT_{ad} . The rate constant combines an Arrhenius term with a smooth ceiling,

$$k_{\text{eff}} = \frac{k_{\text{arr}} k_{\text{max}}}{k_{\text{arr}} + k_{\text{max}}}, \quad k_{\text{arr}} = A e^{-E_a/RT}. \quad (4)$$

With $E_a = 169 \text{ kJ mol}^{-1}$ and $A = 7.3 \times 10^{16} \text{ s}^{-1}$, k_{arr} is about 10^{-4} s^{-1} at 150°C (slow onset) and about 1 s^{-1} at 250°C . The ceiling $k_{\text{max}} = 1 \text{ s}^{-1}$ leaves onset unchanged; its purpose is explained in Section 4.1.

2.3 Barrier designs

Four designs are compared: no barrier (cells in direct contact); a static conductor ($k = 5 \text{ W m}^{-1} \text{ K}^{-1}$); a static, aerogel-like insulator ($k = 0.03 \text{ W m}^{-1} \text{ K}^{-1}$); and a switchable barrier that conducts below a switch temperature T_{sw} and insulates above it. The switch law is a logistic blend in $\ln k$,

$$\ln k(T) = \ln k_{\text{off}} + \frac{\ln(k_{\text{on}}/k_{\text{off}})}{1 + e^{(T-T_{\text{sw}})/w}}, \quad (5)$$

with $k_{\text{on}} = 5 \text{ W m}^{-1} \text{ K}^{-1}$, $k_{\text{off}} = 0.05 \text{ W m}^{-1} \text{ K}^{-1}$, $T_{\text{sw}} = 120^\circ\text{C}$ and $w = 5 \text{ K}$. The switch is reversible and local: every barrier node switches on its own temperature, so a barrier with a steep internal gradient can be insulating on its hot side and conducting on its cold side. Blending $\ln k$ rather than k matters for a $100\times$ switch ratio. A linear blend leaves k several times k_{off} until about 20 K above T_{sw} , so the barrier would barely insulate near its nominal switch point. All barriers share the same volumetric heat capacity, which isolates the effect of conductivity.

2.4 Scenarios and parameters

Abuse. Cell 1 starts at 300°C and everything else at 25°C . A cell is counted as *propagated* when its volume-mean α exceeds 0.9.

Normal operation. Every cell generates a uniform $q_{\text{gen}} = 2 \times 10^4 \text{ W m}^{-3}$, the reaction is switched off (its rate is negligible below 70°C), and Eq. (1) is solved to steady state. The quantity of interest is the peak cell temperature.

Parameters. Table 1 lists the parameters. Two were changed from the initial estimates during development. First, the end cooling coefficient was raised to $h = 100 \text{ W m}^{-2} \text{ K}^{-1}$, representing an end plate connected to a cooling system. With natural-convection values ($h \lesssim 30 \text{ W m}^{-2} \text{ K}^{-1}$) no barrier, not even the insulator, prevented propagation in this 1D model, because the stack ends are the only heat sink and the trigger cell's roughly 20 MJ m^{-2} must eventually leave somewhere. Second, the rate ceiling k_{max} was added (Section 4.1).

Table 1: Model parameters. All values are illustrative order-of-magnitude placeholders and have not been fitted to a specific cell or material.

Quantity	Value	Note
Cell k , ρ , c_p	$0.8 \text{ W m}^{-1} \text{ K}^{-1}$, 2500 kg m^{-3} , $1000 \text{ J kg}^{-1} \text{ K}^{-1}$	through-plane, homogenised single-step kinetics
E_a , A	169 kJ mol^{-1} , $7.3 \times 10^{16} \text{ s}^{-1}$	
ΔT_{ad} , k_{max}	600 K , 1 s^{-1}	
Conductor / insulator k	$5 / 0.03 \text{ W m}^{-1} \text{ K}^{-1}$	static
Switch k_{on} , k_{off} , T_{sw} , w	5 and $0.05 \text{ W m}^{-1} \text{ K}^{-1}$, 120°C , 5 K	reversible
Barrier ρc_p , thickness	$10^6 \text{ J m}^{-3} \text{ K}^{-1}$, 2 mm	all designs
h , T_∞	$100 \text{ W m}^{-2} \text{ K}^{-1}$, 25°C	both ends
q_{gen} (normal operation)	$2 \times 10^4 \text{ W m}^{-3}$	cells only

3 Numerical method

Spatial discretisation. Each layer is meshed separately (40 volumes per cell, 12 per barrier), so every material interface lies on a control-volume face. The face conductance is the series resistance of the two adjoining half-volumes,

$$G_{i+1/2} = \left(\frac{\Delta x_i}{2k_i} + \frac{\Delta x_{i+1}}{2k_{i+1}} \right)^{-1}, \quad (6)$$

which is the distance-weighted harmonic mean of k [20]. It is exact for steady conduction through layers in series and handles the $27\times$ jump at an insulator–cell interface without special treatment. Boundary faces add $1/h$ in series.

Time integration. Reaction and conduction are coupled by first-order (Lie) operator splitting [21]. In the *reaction step*, Eq. (3) is linear in α for a frozen rate constant and has the exact solution

$$\alpha^{n+1} = 1 - (1 - \alpha^n) e^{-k_{\text{eff}} \Delta t}, \quad (7)$$

where k_{eff} is evaluated at a predicted mid-step temperature (an exponential midpoint rule). The released heat is added as $\Delta T_{\text{ad}} (\alpha^{n+1} - \alpha^n)$. The update is stable for any Δt , keeps $0 \leq \alpha \leq 1$ and conserves energy exactly, which makes the stiff kinetics tractable. In the *conduction step*, backward Euler is used with $k(T)$ lagged from the start of the step (Picard refinement is available). The resulting tridiagonal system is solved with a banded direct solver.

Step control. The time step is

$$\Delta t = \min \left(\Delta t_{\text{max}}, \frac{\Delta T_{\text{rxn,max}}}{\max_i \dot{T}_{\text{rxn},i}} \right), \quad (8)$$

with $\Delta T_{\text{rxn,max}} = 2 \text{ K}$ and $\Delta t_{\text{max}} = 0.5 \text{ s}$. This is an accuracy limit rather than a stability limit: it shrinks the step only while some node is igniting.

Steady state. Normal operation is solved by pseudo-transient continuation: backward-Euler steps from ambient with a geometrically growing step. For the switchable barrier this follows the physical start-up path, so if self-heating trips the switch, the solution lands on the hot (insulating) branch.

Energy accounting. The solver tracks stored thermal energy, unreleased chemical energy and cumulative boundary loss. Because both sub-steps are conservative, the discrete energy balance closes to round-off. The largest relative residual in any run reported here is 5×10^{-12} .

4 Verification

The test suite runs automatically on every commit and checks the following.

1. **Energy conservation.** With insulated ends and no reaction, total thermal energy stays constant to a relative 10^{-12} over 2000 s, with the nonlinear switchable barrier in place. With the reaction on, thermal plus chemical energy is conserved, and with convection the budget closes to below 10^{-10} .

2. **Series conduction.** For a cell | barrier | cell stack held between two different ambient temperatures, the steady solution matches the analytical series-resistance profile to 10^{-8} K, for both $6\times$ and $27\times$ conductivity jumps, on a deliberately coarse mesh. For uniform generation in a single cell, the peak matches $T_\infty + qL_c/(2h) + qL_c^2/(8k)$.
3. **Reaction.** A single adiabatic reacting node ends at exactly $T_0 + \Delta T_{\text{ad}}$. Its time to 50 % conversion matches a tight-tolerance implicit ODE solution (Radau IIA [22]) to within 1 %, with and without the rate ceiling.
4. **Convergence.** The cell-2 propagation time converges under mesh refinement with an observed order above 1.5, and under refinement of $\Delta T_{\text{rxn,max}}$ (Fig. 1).

4.1 Why a rate ceiling is needed

The kinetic parameters describe the onset region (150–250 °C). Extrapolated to a burning cell near 900 °C, pure Arrhenius kinetics give $k_{\text{arr}} \sim 10^9 \text{ s}^{-1}$, which implies a reaction front inside the cell of order $\sqrt{\kappa/k_{\text{arr}}} \sim 10^{-7} \text{ m}$, where $\kappa = k/\rho c_p$ is the thermal diffusivity. No practical mesh resolves this, so the front speed is set by the mesh rather than by the physics. The convergence test exposed the problem. Without the ceiling, the cell-2 propagation time drifts from 14.3 to 10.9, 11.1 and 8.8 s as the mesh is refined from 10 to 80 volumes per cell, with no sign of settling (Fig. 1).

Real high-temperature decomposition is limited by transport and multi-step chemistry, not by an ever-faster single step. The smooth ceiling in Eq. (4) with $k_{\text{max}} = 1 \text{ s}^{-1}$ makes a single cell burn over a few seconds, broadens the front to about $\sqrt{\kappa/k_{\text{max}}} \approx 0.6 \text{ mm}$ and restores convergence. The default mesh is within 2 % of a $4\times$ finer one, and halving $\Delta T_{\text{rxn,max}}$ changes the result by 0.1 %. The contained cases are also converged: the peak temperature in cell 2 behind the switchable barrier varies by less than 0.1 K over a $100\times$ range of Δt_{max} , and by about 1 K from 20 to 80 volumes per cell.

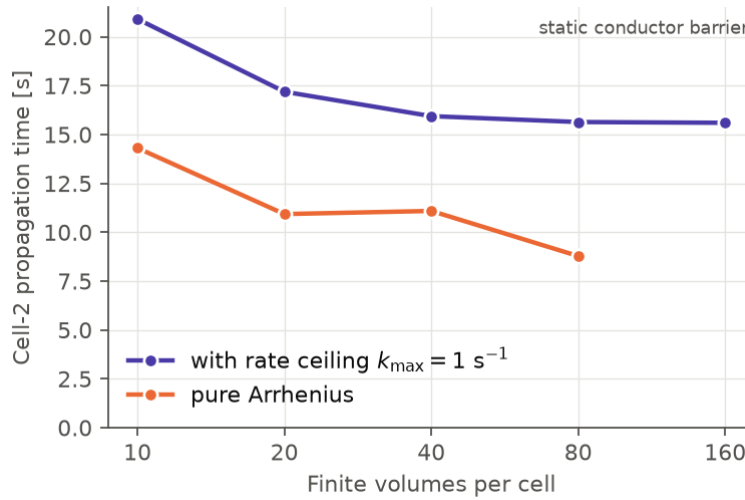


Figure 1: Mesh convergence of the cell-2 propagation time for the static-conductor stack. With pure Arrhenius kinetics the result does not converge, because the reaction front is unresolvably thin. With the rate ceiling it converges at roughly second order.

5 Results

5.1 The tradeoff

Table 2 and Fig. 2 summarise the four designs. Without a barrier or with the conductor, runaway spreads to every cell: each neighbour ignites 14–16 s after the previous one, and the whole stack has reacted within about one minute (Fig. 3). The insulator confines runaway to the trigger cell but raises the normal-operation peak by 26.5 K. The switchable barrier gives the conductor’s 38.0 °C in normal use *and* the insulator’s containment.

Table 2: Outcomes for the four designs (2 mm barriers where present).

Design	Normal-op. peak	Cells in runaway	Propagation times, cells 1–5 [s]
None (direct contact)	37.8 °C	5/5	2.3, 14.1, 28.2, 42.2, 56.1
Static conductor	38.0 °C	5/5	2.3, 15.9, 31.8, 47.8, 63.6
Static insulator	64.5 °C	1/5	2.3, –, –, –, –
Switchable	38.0 °C	1/5	2.3, –, –, –, –

The barrier tradeoff: only the switchable design wins on both

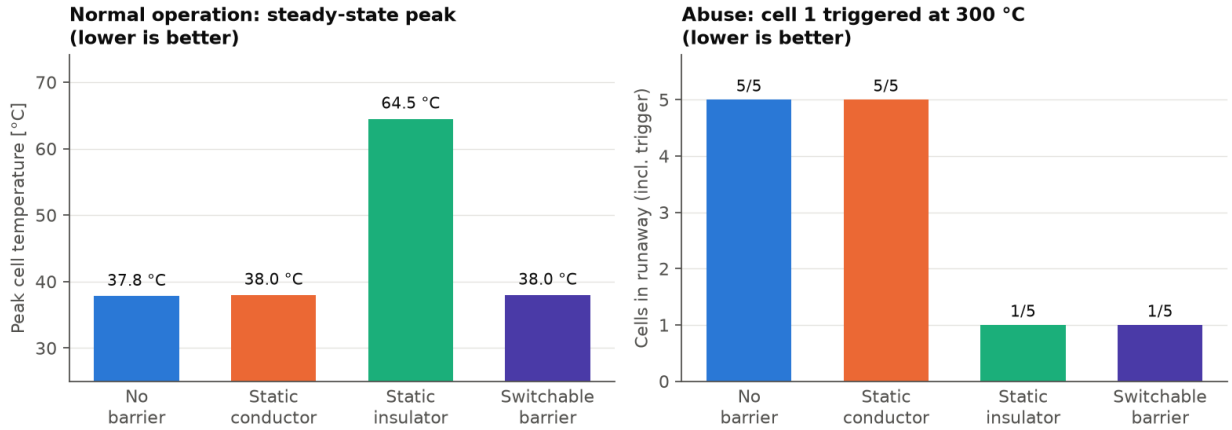


Figure 2: Normal-operation steady-state peak temperature (left) and number of cells in runaway after triggering cell 1 (right) for each design.

The trigger cell peaks at 850–870 °C, slightly below $300 + \Delta T_{\text{ad}} = 900$ °C, because it loses heat while it reacts. Propagated cells settle near $25 + 600$ °C plus whatever preheating they received. Both values are consistent with the energy content of the model.

5.2 Mechanism of containment

Figure 4 shows how the switch works. The hot face of the first barrier switches off and holds a temperature drop of roughly 700 K across a few millimetres while cell 1 sheds its heat through the cooled end. The cold side of the same barrier stays below T_{sw} and keeps conducting, as do the downstream barriers, so cell 2 spreads whatever heat leaks through into cells 3–5. One consequence is easy to miss. Cell 2 reaches a higher local peak behind the switchable barrier (153 °C, 3 % conversion) than behind the static insulator (124 °C, 0.2 %), because only part of the switchable barrier’s thickness is in the insulating state.

Thermal runaway propagation: cell temperatures by barrier design

Cell 1 triggered at 300 °C. Dots mark the time each cell reaches 90 % mean conversion.

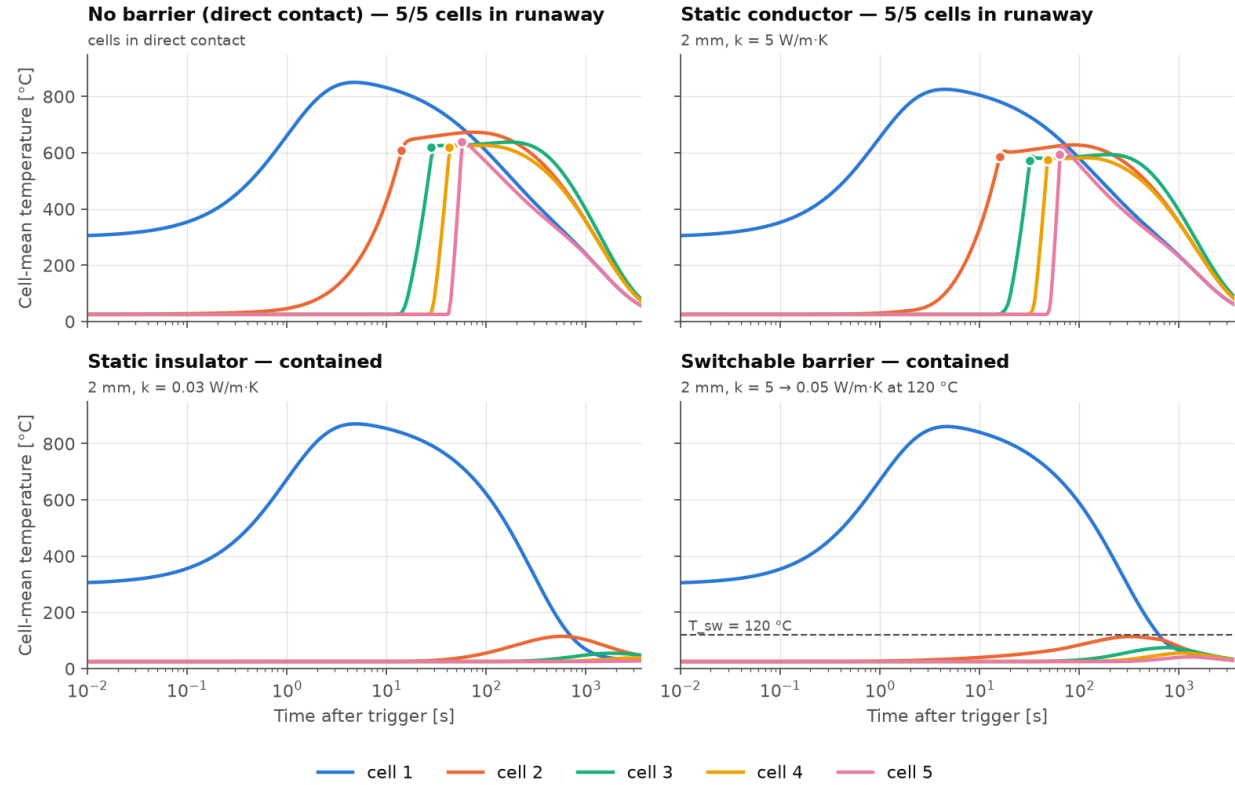


Figure 3: Cell-mean temperatures after triggering cell 1, one panel per barrier design. Dots mark the time each cell reaches 90 % mean conversion.

Where the heat goes: temperature through the stack over time (barriers shaded)

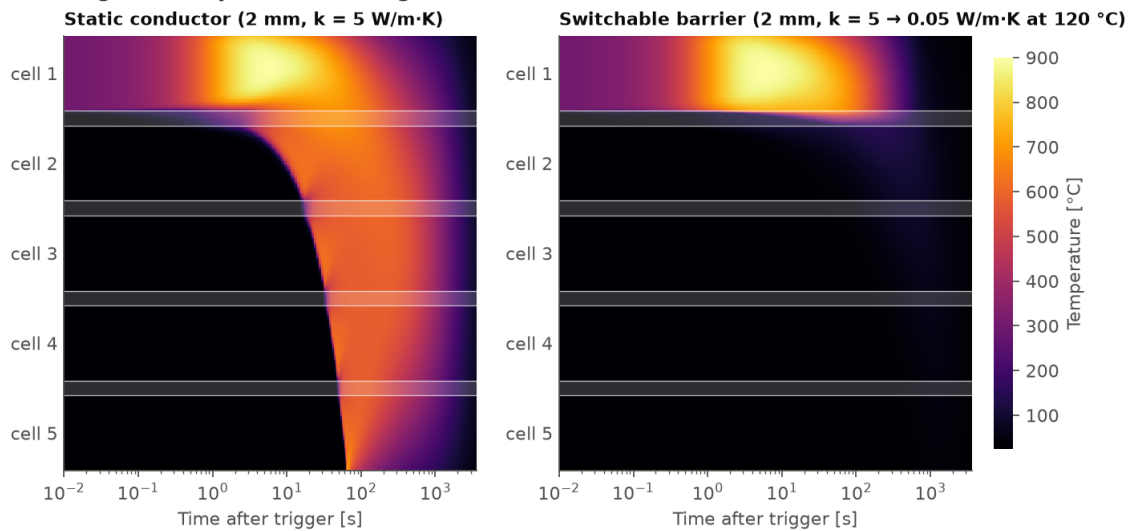


Figure 4: Temperature through the stack over time for the conductor (left) and the switchable barrier (right). Barriers are shaded.

5.3 Design map

Figure 5 varies the switchable barrier’s thickness L (0.5–5 mm) and hot-state conductivity k_{off} ($0.02\text{--}1\text{ W m}^{-1}\text{K}^{-1}$) over 361 simulations. Two features stand out. First, the outcomes are binary: every run ends with either one or all five cells in runaway. Once cell 2 ignites, each later cell faces a trigger at least as severe, so the barrier must stop the first hand-off. Second, the boundary follows a line of constant hot-state conductance, $k_{\text{off}}/L \approx 32\text{ W m}^{-2}\text{K}^{-1}$, about a third of the end-cooling coefficient. Physically, the runaway cell must lose heat to the cooled end faster than it leaks into its neighbour. The default design ($k_{\text{off}}/L = 25\text{ W m}^{-2}\text{K}^{-1}$) lies about 20 % inside the boundary, a modest margin.

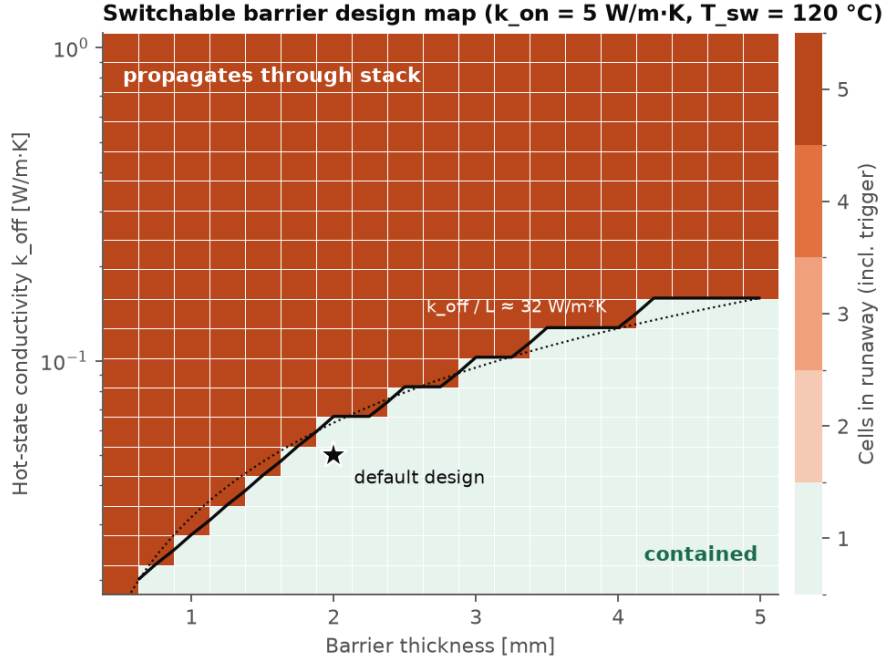


Figure 5: Cells in runaway over barrier thickness and hot-state conductivity ($k_{\text{on}} = 5\text{ W m}^{-1}\text{K}^{-1}$, $T_{\text{sw}} = 120\text{ }^{\circ}\text{C}$). The dotted line is constant k_{off}/L ; the star marks the default design.

5.4 Choosing the switch temperature

Figure 6 sweeps T_{sw} from 25 to $250\text{ }^{\circ}\text{C}$. Below about $38\text{ }^{\circ}\text{C}$ the barrier partly switches off during normal operation and the steady peak rises toward the insulator’s. Above about $175\text{ }^{\circ}\text{C}$ the barrier switches only after the face of cell 2 has passed self-heating onset, and runaway propagates. Between these limits the switch both cools and protects. The lower edge is set by the operating temperature plus a few switch widths; the upper edge is set by the kinetics.

6 Discussion and limitations

The model is deliberately minimal, and several simplifications bound what it can say.

- **One dimension.** Heat flows only through the stack thickness, and the stack ends are the only heat sink. Real packs also lose heat sideways to cold plates, housings and busbars. This is why the results depend strongly on h . The containment threshold $k_{\text{off}}/L \approx h/3$ should be

Choosing the switch temperature

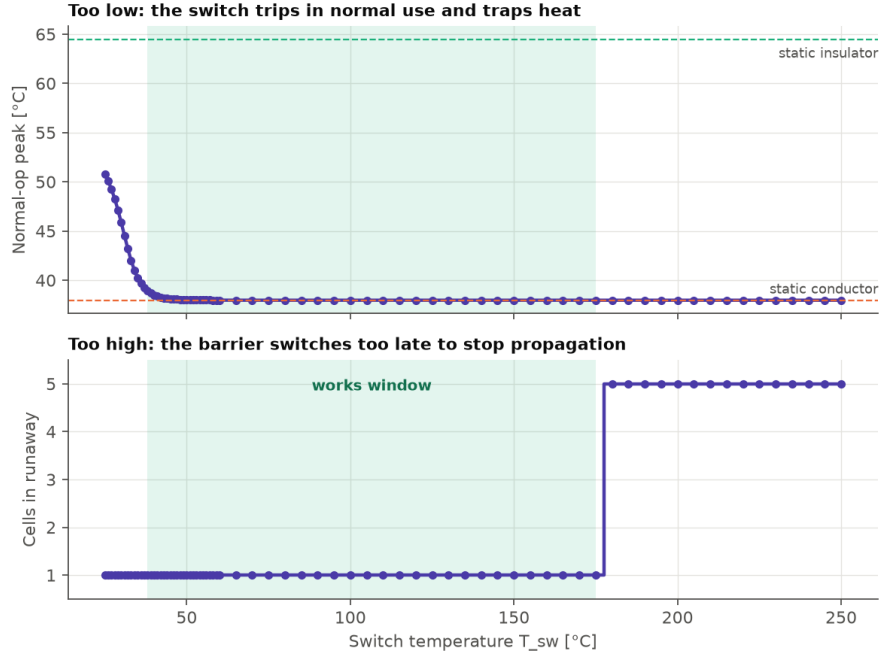


Figure 6: Normal-operation peak (top) and cells in runaway (bottom) as functions of the switch temperature. The shaded window both cools and protects.

read as a statement about the competition between barrier leakage and heat removal, not as a design rule.

- **Lumped kinetics.** One Arrhenius step with a rate ceiling captures onset and total heat release, but not the sequence of real decomposition reactions (SEI breakdown, anode–electrolyte, cathode decomposition, electrolyte combustion, internal short) [1, 10].
- **Missing transport.** Venting, ejecta, gas flow and flames, which often carry much of the energy in real failures [2], are not modelled, and neither are radiation across gaps or contact resistance.
- **An idealised switch.** Switching is instantaneous, reversible and free of hysteresis, degradation and latent heat.
- **Illustrative parameters.** No value is fitted to data, so temperatures and times are qualitative.

Natural next steps are to fit $k(T)$ to measured data for a real switchable material, including hysteresis; to fit multi-step kinetics to accelerating-rate calorimetry and calibrate k_{max} against measured runaway duration; to extend the model to 2D or 3D with lateral cooling; to compare against an independent commercial finite-element model; and to quantify how parameter uncertainty moves the containment boundary.

7 Conclusions

A verified 1D finite-volume model reproduces the barrier conflict between normal-use cooling and abuse containment, and shows that an idealised switchable barrier resolves it. Containment is controlled by the hot-state barrier conductance relative to the heat removal available to the runaway cell. The switch temperature has a broad working window between the normal operating

temperature and the onset of self-heating. On the numerical side, mesh-convergence testing exposed a modelling error that a single run would hide: onset-fitted Arrhenius kinetics extrapolated to burning temperatures make propagation times mesh-dependent, and a physically motivated rate ceiling is needed for a converged answer.

Code and data availability. The model, tests and scripts that regenerate every figure and number in this report are available at <https://github.com/adityameenak/switchable-barrier-runaway> under the MIT license. The solver is written in Python using NumPy [23] and SciPy [24].

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